

Programming Exercise and Assignment: 2

There are MATLAB/Octave programming exercises in this assignment. Submit your solutions as a single zip file with the name <Roll_No>.zip on Moodle.

Programming:

1. Write a program to perform the following linear transformations on a circle with center at 1,3 and a radius of 5 units:
 - a. Rotation about origin by a fixed angle Theta =55 degree
 - b. Reflection about the line $y=2x$
 - c. Stretch the circle by 2 times along the x-axis
2. A plane $z=2x+3y$ exists in 3D Space. Write a program to plot the reflection of the circle with center at [1,3,0] and radius of 5 units about this plane.
3. Write a MATLAB/Octave function that takes a matrix A as input and performs the following:
 - a. Implement the **Gram-Schmidt process** from scratch (without using `orth` or `qr`) to produce an orthonormal basis Q for the column space of A . Verify orthonormality by checking that $Q^T Q = I$.
 - b. Compute the upper triangular matrix R such that $A=QR$, and verify the decomposition by reconstructing A . (note: Do not calculate it using `inv`)
 - c. Use `timeit` to compare the runtime of your implementation against MATLAB's built-in `qr(A)` for a random 50×10 matrix generated with `rand(50, 10)`.
 - d. Using your QR decomposition, solve the least squares problem $Ax=b$ for a random vector b , and verify your answer against MATLAB's built-in `\` operator. Report the norm of the residual in both cases. Generate plots to show the data and the best linear fit.
4. Consider a set of m data points (t_i, y_i) obtained from a mechanical system (e.g., a damped oscillation or stress-strain curve).
 - a. Write a MATLAB function that constructs the **Vandermonde matrix** A for polynomial fitting of degree n , where each row is of the form $[1, t_i, t_i^2, \dots, t_i^n]$.
 - b. Implement the least squares solution using the **normal equations** $A^T A \hat{x} = A^T b$ from scratch (without using `\` or `lsqlin`), and find the best-fit polynomial coefficients.
 - c. Compute and plot the **projection** $\hat{b} = A \hat{x}$ against the original data points, and separately plot the **residual vector** $r = b - \hat{b}$.

- d. Verify numerically that the residual is orthogonal to every column of A (i.e., $\mathbf{r} \approx \mathbf{0}$). Use the following synthetic dataset to test your code:
 $\mathbf{t}=[0, 1, 2, 3, 4, 5], \mathbf{y}=[2.1, 3.9, 6.2, 7.8, 10.1, 12.3]$
- e. Fit polynomials of degree 1 and 2, and comment on which fits better using the norm of the residual.

Theory:

1. Let $P = A(A^T A)^{-1} A^T$ be the projection matrix onto the column space of A , where $A \in \mathbb{R}^{m \times n}$ has linearly independent columns. Prove that:
 - a. P is symmetric: $P^T = P$.
 - b. P is idempotent: $P^2 = P$.
 - c. $I - P$ is also a projection matrix, and identify the subspace it projects onto.
 - d. The eigenvalues of P are only 0 and 1.
2. Given an inconsistent system $A\mathbf{x} = \mathbf{b}$ where $A \in \mathbb{R}^{m \times n}$, $m > n$:
 - a. Prove that the least squares solution $\hat{\mathbf{x}}$ satisfies the normal equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.
 - b. Prove that the residual $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to the column space of A .
 - c. If A has full column rank, prove that $A^T A$ is symmetric positive definite, and hence the least squares solution is unique.

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